Module 6: Overall Final Project

Hostel Portal Project | Student Project Report

1. Aim

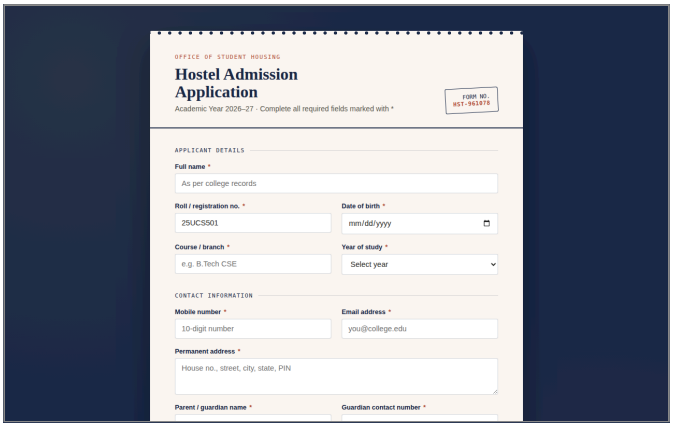
To combine all modules into one working project: **Hostel Portal - Login + Application (Registration) + Validation + Database**.

2. Description

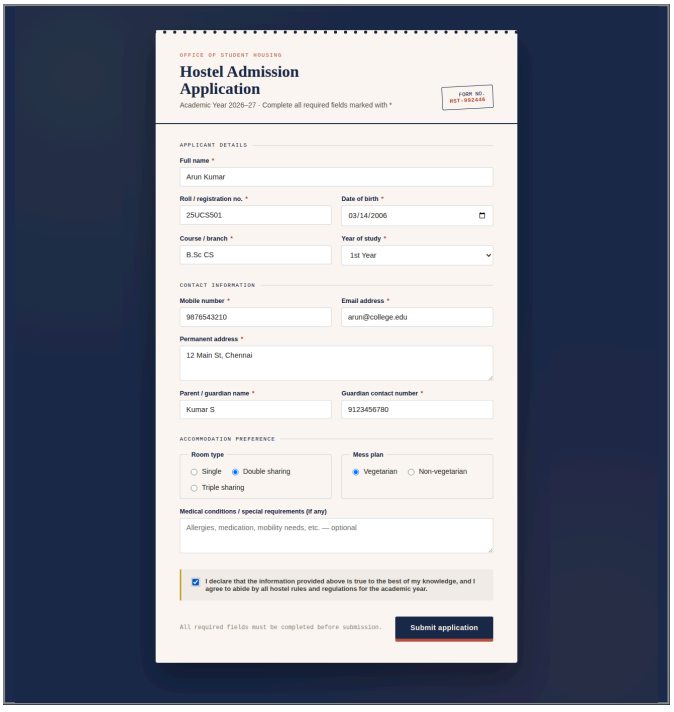
The project is one HTML file (`hostel_portal.html`). A student logs in with a roll number (25ucs501 - 25ucs565), fills the Hostel Admission Application for 2026-27, and submits it. The script validates the data, then shows a Form No. and Reference ID as confirmation. The database design (Module 5) holds the same fields, ready to store each application.

3. Working of the Project (screenshots)

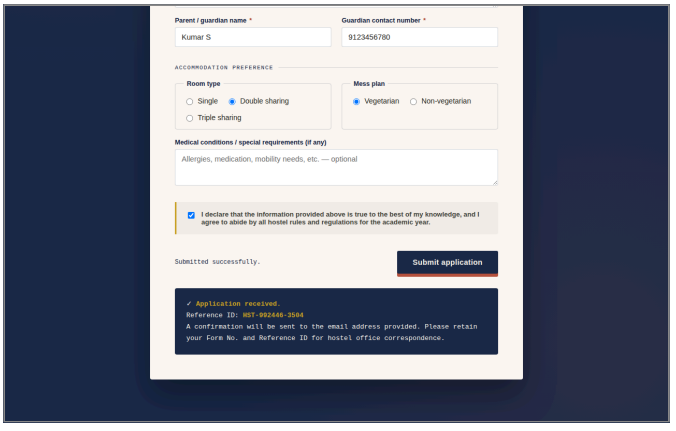




Step 2: Application form after login (Roll No. auto-filled)



Step 3: All fields filled, declaration ticked



Step 4: Submitted - Reference ID shown

Sample output: Form No. **HST-992446** → Reference ID **HST-992446-3504** (a new random number is generated every time).

4. How the Modules Are Joined

Module	Role in the final project	Where in the code
1. Bio data (HTML)	HTML structure, tables, CSS styling used to design the pages	HTML + <style>
2. Login page	Entry point; only 25ucs501-565 with password 12345678	`#loginScreen`, `#loginForm`
3. Registration page	Hostel application form: 3 sections + declaration	`#hostelForm`
4. Validation scripting	Checks login and form data before submit	<script> section
5. Database tables	students and hostel_applications store the data	module5_database.sql

5. Program Flow

> Open page → Login screen → Enter Roll No. + Password → **Valid?** No: error message / Yes: show form (Roll No. filled, Form No. generated) → Fill details → Submit → **All fields valid?** No: red message, stay on form / Yes: Reference ID + success box → (server inserts the row into hostel_applications)

6. How to Run

1. Save **hostel_portal.html** on the computer and double-click it (any browser).
2. Enter User ID **25ucs501** and password **12345678**; click Sign In.
3. Fill every field marked *, choose room type and mess plan, tick the declaration.
4. Click **Submit application**. Wrong or missing data shows a message; correct data shows the Reference ID.

7. Final Testing Summary

Area	Tests	Result
Login (valid and invalid IDs, range, password, spaces, capitals)	8	All passed
Application form (empty, mobile, email, full valid submit)	6	All passed
Database (constraints: CHECK, FOREIGN KEY, UNIQUE, valid insert)	5	All passed
Total	19	All 19 passed

8. Project Summary

Item	Details
Project name	Hostel Portal - Login & Application
Technologies	HTML5, CSS3, JavaScript, SQL (SQLite / MySQL)
Users	Students with roll numbers 25ucs501 - 25ucs565
Output	Form No., Reference ID, confirmation message
Limitations	Login is checked in the browser only; the submitted data is not yet saved to the database automatically
Future improvements	Server-side login with hashed passwords, automatic saving to the database, e-mail confirmation, admin page to allot rooms

9. Complete Source Code: hostel_portal.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Hostel Portal — Login & Application</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link href="https://fonts.googleapis.com/css2?
family=Fraunces:ital,wght@0,500;0,600;0,700;1,500&family=Inter:wght@400;500;600;700&family=JetBrains+Mono:wght@400;600&display=swap"
rel="stylesheet">
<style>
  :root{--navy:#1B2A4A;--cream:#F5F0E6;--brick:#B5533C;--gold:#C9A227;--ink:#2E2E2A;--paper:#FBF8F1;--line:rgba(27,42,74,0.18)}
  *{box-sizing:border-box} html,body{margin:0;padding:0;min-height:100vh}
  body{font-family:'Inter',sans-serif;color:var(--ink);background:var(--navy)}
  .login-screen{position:fixed;inset:0;width:100%;height:100vh;background:linear-gradient(to
right,rgba(27,42,74,.3),rgba(27,42,74,.65)),url('https://images.unsplash.com/photo-1562774053-701939374585?q=80&w=1920&auto=format&fit=crop')
center/cover no-repeat;display:flex;align-items:center;justify-content:flex-end;padding:24px 8%;z-index:100;transition:opacity .4s ease,visibility
.4s ease}
  .login-screen.hidden{opacity:0;visibility:hidden;pointer-events:none}
  .login-card{width:100%;max-width:380px;background:rgba(251,248,241,.92);backdrop-filter:blur(12px);-webkit-backdrop-filter:blur(12px);border:1px
solid rgba(255,255,255,.4);border-radius:8px;padding:36px 32px;box-shadow:0 25px 50px -12px rgba(0,0,0,.5)}
  .login-header{margin-bottom:24px}.login-header .eyebrow,.brand-block .eyebrow{font-family:'JetBrains Mono',monospace;font-size:10px;letter-
spacing:.14em;text-transform:uppercase;color:var(--brick);margin:0 0 6px}
  .login-header h2{font-family:'Fraunces',serif;font-weight:700;font-size:24px;color:var(--navy);margin:0 0 6px}.login-header p{margin:0;font-
size:12.5px;color:#5b5b52}
  .login-form .field{margin-bottom:18px;display:flex;flex-direction:column;gap:6px}.login-form label{font-size:12px;font-weight:600;color:var(--
navy)}
  .login-form input{font-family:'Inter',sans-serif;font-size:14px;color:var(--ink);background:#fff;border:1.5px solid var(--line);border-
radius:4px;padding:10px 12px;outline:none}
  .login-form input:focus{border-color:var(--navy);box-shadow:0 0 0 3px rgba(27,42,74,.12)}
  .login-btn{width:100%;margin-top:10px;font-family:'Inter',sans-serif;font-weight:700;font-size:14px;color:var(--cream);background:var(--
navy);border:none;border-radius:4px;padding:12px;cursor:pointer;box-shadow:0 4px 0 var(--brick)}
  .login-btn:hover{background:#233657}.login-btn:active{transform:translateY(2px);box-shadow:0 2px 0 var(--brick)}
  .error-msg{display:none;margin-top:16px;padding:10px 12px;background:rgba(181,83,60,.1);border-left:3px solid var(--brick);color:var(--brick);font-
size:12px;font-weight:500;border-radius:0 3px 3px 0}.error-msg.visible{display:block}
  .app-container{display:none;padding:48px 16px 80px;justify-content:center;background:var(--navy);background-image:radial-gradient(circle at 10%
10%,rgba(201,162,39,.08),transparent 40%),radial-gradient(circle at 90% 90%,rgba(181,83,60,.1),transparent 40%);min-height:100vh}.app-
container.visible{display:flex}
  .sheet{width:100%;max-width:720px;background:var(--paper);border-radius:4px;box-shadow:0 30px 60px
rgba(0,0,0,.35);position:relative;overflow:hidden}
  .sheet::before{content:"";position:absolute;top:0;left:0;right:0;height:10px;background-image:radial-gradient(circle,var(--navy) 3px,transparent
3.5px);background-size:20px 20px;background-position:8px -6px;background-color:var(--paper)}
  header.form-head{padding:44px 48px 28px;border-bottom:2px solid var(--navy);display:flex;justify-content:space-between;align-items:flex-
end;gap:20px}
  .brand-block .eyebrow{font-size:11px;margin:0 0 10px}.brand-block h1{font-family:'Fraunces',serif;font-weight:700;font-size:32px;line-
height:1.05;margin:0 0 8px;color:var(--navy)}.brand-block p{margin:0;font-size:14px;color:#5b5b52}
  .stamp{font-family:'JetBrains Mono',monospace;font-size:11px;text-align:right;color:var(--navy);border:1.5px solid var(--navy);border-
radius:3px;padding:10px 14px;transform:rotate(-3deg);white-space:nowrap}.stamp span{display:block;color:var(--brick);font-weight:600;letter-
spacing:.05em}
  form{padding:36px 48px 48px}.section-label{font-family:'JetBrains Mono',monospace;font-size:11px;letter-spacing:.12em;text-
transform:uppercase;color:var(--navy);display:flex;align-items:center;gap:10px;margin:34px 0 18px}.section-label:first-of-type{margin-top:0}.section-
label::after{content:"";flex:1;height:1px;background:var(--line)}
  .grid{display:grid;grid-template-columns:1fr 1fr;gap:18px 20px}.field{display:flex;flex-direction:column;gap:6px}.field.span-2{grid-column:1/-1}
  label{font-size:12.5px;font-weight:600;color:var(--navy)}label .req{color:var(--brick);margin-left:2px}
  input,select,textarea{font-family:'Inter',sans-serif;font-size:14.5px;color:var(--ink);background:#fff;border:1.5px solid var(--line);border-
radius:3px;padding:10px 12px;outline:none}input:focus,select:focus,textarea:focus{border-color:var(--navy);box-shadow:0 0 0 3px
rgba(27,42,74,.1)}textarea{resize:vertical;min-height:70px}
  fieldset{border:1.5px solid var(--line);border-radius:4px;padding:14px 16px;margin:0}fieldset legend{font-size:12.5px;font-weight:600;color:var(--
navy);padding:0 6px}.radio-row{display:flex;flex-wrap:wrap;gap:16px;margin-top:4px}.radio-row label{display:flex;align-items:center;gap:6px;font-
weight:500;font-size:14px;color:var(--ink)}.radio-row input{width:auto;padding:0}
  .declaration{margin-top:32px;padding:16px 18px;background:rgba(27,42,74,.045);border-left:3px solid var(--gold);border-radius:0 4px 4px 0;font-
size:13px;color:#4a4a42;display:flex;gap:10px;align-items:flex-start}.declaration input{margin-top:3px;width:auto}
  .actions{margin-top:32px;display:flex;align-items:center;justify-content:space-between;gap:16px;flex-wrap:wrap}.hint{font-
size:12px;color:#8a8a7d;font-family:'JetBrains Mono',monospace}
  button.submit{font-family:'Inter',sans-serif;font-weight:700;font-size:14.5px;letter-spacing:.02em;color:var(--cream);background:var(--
navy);border:none;border-radius:3px;padding:14px 30px;cursor:pointer;box-shadow:0 6px 0 var(--
brick)}button.submit:hover{background:#233657}button.submit:active{transform:translateY(4px);box-shadow:0 2px 0 var(--brick)}
  .success{display:none;margin-top:28px;padding:20px 22px;background:#1B2A4A;color:var(--cream);border-radius:4px;font-family:'JetBrains
Mono',monospace;font-size:13px;line-height:1.7}.success.visible{display:block}.success b{color:var(--gold)}
  @media(max-width:640px){.login-screen{justify-content:center;padding:16px}.grid{grid-template-columns:1fr}header.form-head{flex-
direction:column;align-items:flex-start;padding:32px 24px 22px}form{padding:28px 24px 36px}}
</style>
</head>
<body>
<div class="login-screen" id="loginScreen">
  <div class="login-card">
    <div class="login-header"><p class="eyebrow">Student Portal</p><h2>Hostel Sign In</h2><p>Enter your Roll Number & Password to access the
application form.</p></div>
    <form class="login-form" id="loginForm">
      <div class="field"><label for="userId">User ID (Roll No.)</label><input type="text" id="userId" placeholder="e.g. 25ucs501" required></div>
      <div class="field"><label for="password">Password</label><input type="password" id="password" placeholder="••••••••" required></div>
      <button type="submit" class="login-btn">Sign In</button>
      <div class="error-msg" id="loginError">Invalid credentials. Roll No. must be between <b>25ucs501</b> and <b>25ucs565</b> with password
<b>12345678</b>.</div>
    </form>
  </div>
</div>

<div class="app-container" id="appContainer">
  <div class="sheet">
    <header class="form-head">
      <div class="brand-block"><p class="eyebrow">Office of Student Housing</p><h1>Hostel Admission<br>Application</h1><p>Academic Year 2026–27 ·
Complete all required fields marked with *</p></div>
      <div class="stamp">FORM NO.<span id="formNo">HST-000000</span></div>
```

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    </header>
    <form id="hostelForm" novalidate>
      <p class="section-label">Applicant details</p>
      <div class="grid">
        <div class="field span-2"><label>Full name <span class="req">*</span></label><input type="text" name="fullName" required placeholder="As per
college records"></div>
        <div class="field"><label>Roll / registration no. <span class="req">*</span></label><input type="text" name="rollNo" id="formRollNo" required
placeholder="e.g. 25ucs501" readonly></div>
        <div class="field"><label>Date of birth <span class="req">*</span></label><input type="date" name="dob" required></div>
        <div class="field"><label>Course / branch <span class="req">*</span></label><input type="text" name="course" required placeholder="e.g.
B.Tech CSE"></div>
        <div class="field"><label>Year of study <span class="req">*</span></label><select name="year" required><option value="" disabled
selected>Select year</option><option>1st Year</option><option>2nd Year</option><option>3rd Year</option><option>4th Year</option>
<option>Postgraduate</option></select></div>
      </div>
      <p class="section-label">Contact information</p>
      <div class="grid">
        <div class="field"><label>Mobile number <span class="req">*</span></label><input type="tel" name="mobile" required pattern="[0-9]{10}"
placeholder="10-digit number"></div>
        <div class="field"><label>Email address <span class="req">*</span></label><input type="email" name="email" required
placeholder="you@college.edu"></div>
        <div class="field span-2"><label>Permanent address <span class="req">*</span></label><textarea name="address" required placeholder="House
no., street, city, state, PIN"></textarea></div>
        <div class="field"><label>Parent / guardian name <span class="req">*</span></label><input type="text" name="guardianName" required></div>
        <div class="field"><label>Guardian contact number <span class="req">*</span></label><input type="tel" name="guardianMobile" required
pattern="[0-9]{10}" placeholder="10-digit number"></div>
      </div>
      <p class="section-label">Accommodation preference</p>
      <div class="grid">
        <fieldset><legend>Room type</legend><div class="radio-row"><label><input type="radio" name="roomType" value="Single" required> Single</label>
<label><input type="radio" name="roomType" value="Double sharing"> Double sharing</label><label><input type="radio" name="roomType" value="Triple
sharing"> Triple sharing</label></div></fieldset>
        <fieldset><legend>Mess plan</legend><div class="radio-row"><label><input type="radio" name="mess" value="Veg" required> Vegetarian</label>
<label><input type="radio" name="mess" value="Non-veg"> Non-vegetarian</label></div></fieldset>
        <div class="field span-2"><label>Medical conditions / special requirements (if any)</label><textarea name="medical" placeholder="Allergies,
medication, mobility needs, etc. — optional"></textarea></div>
      </div>
      <label class="declaration"><input type="checkbox" name="declaration" required><span>I declare that the information provided above is true to
the best of my knowledge, and I agree to abide by all hostel rules and regulations for the academic year.</span></label>
      <div class="actions"><span class="hint" id="statusHint">All required fields must be completed before submission.</span><button type="submit"
class="submit">Submit application</button></div>
      <div class="success" id="successBox"><div>&#10003; <b>Application received.</b></div><div>Reference ID: <b id="refId"></b></div><div>A
confirmation will be sent to the email address provided. Please retain your Form No. and Reference ID for hostel office correspondence.</div></div>
    </form>
  </div>
</div>

<script>
const
loginForm=document.getElementById('loginForm'),userIdInput=document.getElementById('userId'),passwordInput=document.getElementById('password'),loginE
rror=document.getElementById('loginError'),loginScreen=document.getElementById('loginScreen'),appContainer=document.getElementById('appContainer'),fo
rmRollNo=document.getElementById('formRollNo');
loginForm.addEventListener('submit',function(e){e.preventDefault();const
uid=userIdInput.value.trim().toLowerCase(),pwd=passwordInput.value.trim(),match=uid.match(/^25ucs(\d{3})$/);let isValidUser=false;if(match){const
num=parseInt(match[1],10);if(num>=501&&num<=565)isValidUser=true}if(isValidUser&&pwd==='12345678')
{loginError.classList.remove('visible');formRollNo.value=uid.toUpperCase();loginScreen.classList.add('hidden');appContainer.classList.add('visible')}
else loginError.classList.add('visible')});
const formNoEl=document.getElementById('formNo'),randomFormNo='HST-'+Math.floor(100000+Math.random()*900000);formNoEl.textContent=randomFormNo;
const
form=document.getElementById('hostelForm'),successBox=document.getElementById('successBox'),refId=document.getElementById('refId'),statusHint=documen
t.getElementById('statusHint');
form.addEventListener('submit',function(e){e.preventDefault();if(!form.checkValidity()){form.reportValidity();statusHint.textContent='Please complete
all required fields before submitting.';statusHint.style.color='#B5533C';return}const ref=randomFormNo+'-
'+Math.floor(1000+Math.random()*9000);refId.textContent=ref;successBox.classList.add('visible');statusHint.textContent='Submitted
successfully.';statusHint.style.color='#1B2A4A';successBox.scrollIntoView({behavior:'smooth',block:'nearest'})});
</script>
</body>
</html>
```

10. Conclusion

> All five modules were joined into one working project. The portal allows only registered students to log in, validates the hostel application, confirms the submission with a Form No. and Reference ID, and has a tested database design ready to store the data.